

Ancestral-state reconstruction on a supplied tree

Local configurations or declared repertoires are the modeled states

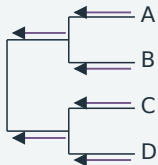
A Tree + tip observations



Tip compatibility

Root, topology and branch lengths are supplied model conditions

B Continuous-time conditional likelihood



Tip observations define compatibility weights

$$\mathbf{m}_c(i) = \sum_j \mathbf{P}_{ij}(t_c) L_c(j)$$

$$L_v(i) = \prod_{c \in \text{children}(v)} \mathbf{m}_c(i)$$

$$\mathcal{L} = \sum_i \pi_i L_{\text{root}}(i)$$

$$\mathbf{P}(t) = \exp(\mathbf{Q}t)$$

Child messages combine by sum-product

Node outputs: marginal ancestral states

Branches: conditional event expectations

C Maximum parsimony



Minimize edit cost; retain ties

D Shared-rate estimation

≥2 independent genes

Pooled likelihood; one scale ρ

Relative rates · roots · trees fixed

Prespecified independent genes

E Assumption sensitivity

Recalculate across declared

root-state distributions

branch lengths or catalogues

Conditional structural summaries

$$q_{ij} = \rho \sum_{e:i \rightarrow j} \lambda_{k(e)} w_e \quad (i \neq j)$$

$$q_{ii} = -\sum_{j \neq i} q_{ij}$$

ρ is one shared scale; λ are fixed relative event rates; w allocates outcomes.

Tip constraints depend on annotation coverage; molecular mechanisms are outside this model.